Chapter 11 – Case Studies (III): Multimedia Problems

IERG4320/IEMS5726 Data Science in Practice | Cheat Sheet

Multimedia Applications

多媒体应用：音频分离、语音合成/识别、图像描述/生成、视频生成

Applications: surveillance, video highlights, healthcare (image diagnosis, drug discovery), accessibility.

Vocal Separation

人声分离：从歌曲中分离人声和伴奏

Frequency Filtering (librosa)

频率过滤：人声频率约80-1100Hz，用滤波器分离

```
import librosa, numpy as np, soundfile as sf
y, sr = librosa.load('song.mp3', duration=120)
S_full, phase = librosa.magphase(librosa.stft(y))
# Nearest-neighbor filter with median
S_filter = librosa.decompose.nn_filter(
  S_full, aggregate=np.median,
  metric='cosine',
  width=int(librosa.time_to_frames(2, sr=sr)))
S_filter = np.minimum(S_full, S_filter)
# Soft masks for vocal/instrument separation
margin_i, margin_v, power = 2, 10, 2
mask_i = librosa.util.softmask(
  S_filter, margin_i*(S_full-S_filter), power=power)
mask_v = librosa.util.softmask(
  S_full-S_filter, margin_v*S_filter, power=power)
# Recover audio
y_vocal = librosa.istft(mask_v * S_full * phase)
y_bg = librosa.istft(mask_i * S_full * phase)
sf.write("vocal.wav", y_vocal, sr)
```

Limitation: removes vocal harmonics & instruments in same range.

Demucs Model

Demucs：U-Net编解码器+LSTM，时域处理

U-Net encoder-decoder with BiLSTM. Time-domain processing.

Input: raw waveform. **Output**: vocals, drums, bass, other.

Encoder: 1→64→128→256→512→1024 channels (4x downsample per layer).

Decoder: symmetric, with skip connections to preserve temporal details.

BiLSTM: between encoder/decoder (hidden=1024, 2–3 layers, dropout 0.2–0.3).

Hybrid Transformer Demucs: evolved from LSTM (context ∼1–2s) to Transformer for global attention. Dual-path: CNN (local features) + Transformer (global context).

混合版：CNN提取局部特征 + Transformer建模全局上下文

Speech Synthesis (TTS)

语音合成：文字转语音

Non-DL Approaches

- **SPSS/HMM**: text normalization → phoneme → HMM generates speech params (Tool: FreeTTS) 统计参数合成
- **Concatenative**: join recorded speech units (phoneme/diphone/syllable). Natural but large storage, join artifacts. 拼接合成：拼接录音片段

WaveNet

WaveNet：自回归卷积模型，逐样本生成音频

Autoregressive: $p(x) = \prod_{t=1}^{T} p(x_t|x_1, ..., x_{t-1})$.

Causal convolution: shifts filter so output only depends on past.

Dilated convolution: skip inputs to increase receptive field exponentially without adding layers.

因果卷积保证时序; 空洞卷积扩大感受野

Softmax: μ-law companding quantizes 65536 values → 256.

Gated activation: $\tanh \otimes \sigma$ (better than ReLU for audio).

Tacotron 2

Tacotron2：端到端TTS，字符→梅尔频谱→WaveNet→波形

Pipeline: characters → mel-spectrograms → WaveNet → waveform.

Encoder: char embedding + conv layers + BiLSTM.

Attention: location-sensitive (prevents skipping/repeating).

Decoder: autoregressive, predicts 80-band mel-spectrogram frames.

MOS (Mean Opinion Score)

MOS：人类主观评分(1-5)，4.3-4.5=优秀，<3.5=不可接受

$\text{MOS} = \frac{1}{N}\sum_{i=1}^{N} R_i$ (1=Bad to 5=Excellent).

WaveNet achieves near-human MOS (∼4.2–4.5).

Speech Recognition (STT)

语音识别：语音转文字

Types: speaker-dependent (robust, trained on one person) vs speaker-independent (general).

Pipeline: digitalize → encoding model (HMM/WaveNet) → decoding language model (Transformer).

Wav2Vec 2.0

Wav2Vec 2.0：CNN编码器+Transformer+量化

CNN encoder (raw audio → latent) → Transformer (long-range context) → quantization (discrete speech units).

Training: masked prediction with contrastive + diversity loss.

HuBERT

HuBERT：用k-means伪标签的自监督语音模型

Hidden-unit BERT: BERT adapted for continuous speech.

Key idea: k-means on MFCCs creates pseudo-labels; model predicts masked audio regions.

Two-stage training:

1. K-means on MFCC features → basic acoustic units.
2. K-means on HuBERT features → linguistic representations.

vs Wav2Vec 2.0: cross-entropy loss (simpler); separate clustering for targets; re-uses encoder embeddings.

Image Captioning

图像描述：给图片自动生成文字描述

Need to identify: objects, actions, attributes, relationships, grammar.

Classic CNN-RNN

经典方法：CNN编码图像 + RNN解码生成文字

Encoder: pre-trained CNN (VGG/ResNet/Inception) → feature vector.

Decoder: LSTM/GRU generates caption word-by-word.

Soft attention: weighted sum over spatial features; $z = \sum p_i \cdot a_i$.

Beam search: find optimal word sequence (not greedy).

AoA (Attention on Attention): information gate + relevance gate for better attention filtering.

Limitation: single feature vector bottleneck loses spatial details.

Transformer-Based

Encoder → ViT (image as patch sequence). Decoder → text Transformer.

ViT把图像切成patch序列，用Transformer编码

CNN Milestones (ImageNet)

ImageNet CNN里程碑

- **AlexNet** (2012): CNNs go mainstream
- **VGGNet** (2014): depth matters
- **GoogLeNet** (2014): Inception module
- **ResNet** (2015): skip connections
- **ViT** (2020): Transformer on images

Captioning Metrics

- **BLEU**: n-gram precision vs references n-gram精确率
- **METEOR**: adds stemming + synonymy (F-measure) 考虑词干和同义词
- **CIDEr**: TF-IDF cosine similarity with references TF-IDF余弦相似度
- **SPICE**: semantic scene graph F-measure 语义场景图

Image Generation

图像生成：从文本/噪声生成图像

Harder than captioning: high-dimensional output ($32\times32 = 3072$ pixels), must imagine background.

GANs

GAN：生成器vs判别器的对抗博弈

Generator creates fake images; Discriminator detects real/-fake.

Minimax objective: $\min_G \max_D \mathbb{E}[\log D(x)] + \mathbb{E}[\log(1 - D(G(z)))]$.

DALL-E

DALL-E：12B参数GPT-3 + dVAE视觉码本

Step 1: dVAE learns visual codebook (image $\rightarrow$ 32×32 tokens).

Step 2: Transformer predicts image tokens from text tokens (256 text + 1024 image tokens, 64 attention layers, 12B params).

CLIP: ranks DALL-E outputs by text-image similarity.

```
from sentence_transformers import SentenceTransformer
from PIL import Image
model = SentenceTransformer("clip-ViT-B-32")
img_emb = model.encode(Image.open("dogs.jpg"))
text_emb = model.encode(["Two dogs in snow",
  "A cat on table", "London at night"])
scores = model.similarity(img_emb, text_emb)
```

Diffusion Model / Stable Diffusion

扩散模型：逐步加噪→学习去噪，从纯噪声生成图像

Forward: add Gaussian noise over T steps (fixed process).

Reverse: neural network learns to denoise step-by-step (Markov chain + KL divergence).

vs Autoencoder: only train one model (denoiser), forward process is fixed.

FID (Fréchet Inception Distance): Wasserstein distance between Inception v3 features of real vs generated images. Lower = better.

FID：用Inception v3特征比较生成图与真实图的分布距离

Video Generation / DeepFake

DeepFake：深度学习+伪造，用AI换脸

Process: collect video $\rightarrow$ encoder learns shared face representation $\rightarrow$ two decoders (one per person) $\rightarrow$ swap: encode A's face, decode with B's decoder.

Loss functions:

Perceptual: $\|\phi_j(x) - \phi_j(G(z))\|^2$ (semantic similarity via VGG).

Identity: $1 - \cos(\phi_{id}(\text{fake}), \phi_{id}(\text{real}))$ (preserve target identity).

How to spot: unnatural movement, irregular blinking, lip sync mismatch, hair/skin artifacts. Best AI detection ~80%.

Ch11 Model Map:
Audio: librosa (filter), Demucs (U-Net+LSTM/Transformer)
TTS: SPSS, Concatenative, WaveNet, Tacotron2
STT: Wav2Vec 2.0, HuBERT
Image Caption: CNN-RNN+Attention, ViT+Transformer
Image Gen: GAN, DALL-E+CLIP, Diffusion
Video Gen: DeepFake (encoder-decoder swap)